

FIRST-ORDER MATCHING AND PRECISION COSTS IN CONTRACTING CONTROL SYSTEMS

AUTHOR INFORMATION PENDING

ABSTRACT. We study the minimum number of open-loop inputs needed to keep each actual initial state within a Euclidean tolerance of a triangular constraint at every time of a finite horizon. Two planar C^2 diffeomorphisms have a common contracting origin and make the constraint controlled invariant. Their first derivatives have limiting angular branch proportions a_0, a_1 and radial factors $r_i = a_i^{\beta_i}$, where $\beta_i > 1$. Within this specified first-order family, all positive-area compact initial sets share a cost rate for sufficiently small positive precision exponents if and only if $\beta_0 = \beta_1$. Under mismatch, the full-constraint rate is γD , whereas one fixed compact set of arbitrarily nearly full area has rate $\gamma h / \chi_R$, strictly smaller, for $0 < \gamma < -\log \max_i r_i$. Here $r_0^D + r_1^D = 1$, $h = -\sum_i a_i \log a_i$ and $\chi_R = -\sum_i a_i \log r_i$. The same set works for every tolerance sequence $\delta_n = \exp(-\gamma n + o(n))$ in this range. Under matching, $\beta_i = \beta$, we determine the full spectrum $\max_p H_b(p) \min\{1, \gamma / [\beta \chi(p)]\}$ for all finite $\gamma \geq 0$, where H_b is the binary entropy and $\chi(p) = -p \log a_0 - (1-p) \log a_1$. Its common-cost and saturation boundaries are βh and $-(\beta/2) \log(a_0 a_1)$. In that case a single nearly full-area compact set has rate $\min\{h, \gamma/\beta\}$ for all exponents and sequences. The proof separates angular and radial products in an actual-area bound for every approximate input, including exit and reentry. Complete feasible intervals may overlap or have empty or truncated images. Physical bounded-run trajectories provide the lower bounds; stopping scales and type counting supply the corresponding upper bounds.

1. INTRODUCTION

Invariance entropy measures the number of control signals needed to maintain a state constraint; see Kawan [8]. When every input contracts toward a common point, a finite catalogue can suffice for arbitrarily long horizons at each fixed positive tolerance. The catalogue can nevertheless grow exponentially when the tolerance decreases exponentially with the horizon. We study this joint limit and ask whether its cost is shared by all compact initial sets of positive area.

The distinction between typical initial states and the full constraint is essential. A compact set of almost full area can omit rare states whose required controls determine the full-set cost. At the same time, an approximately feasible input need not follow a selected exact code. It may switch between overlapping feasible branches, leave the constraint within the permitted tolerance, and later return. A symbolic count alone does not settle the minimum physical catalogue.

The main question is whether the angular branch geometry and radial contraction must match for a common low-precision cost to exist. Theorem 2.1 answers this within a specified first-order family, under controlled feasibility of the actual triangle, global invertibility, and a common norm inequality toward the origin. The radial factors are $r_i = a_i^{\beta_i}$. The matching condition $\beta_0 = \beta_1$ is necessary and sufficient. No radial autonomy, boundary-ray preservation, globally small derivatives, compact support, or full angular image for every feasible word is required.

2020 *Mathematics Subject Classification.* Primary 93C55, 37B40; Secondary 93D15, 37C05.

Key words and phrases. Outer invariance complexity, first-order matching, precision spectrum, controlled contraction, initial-set dependence.

When the powers differ, Theorem 2.2 separates the full-constraint cost γD from $\gamma h/\chi_R$ on a single nearly full-area compact initial set, for every sufficiently small positive exponent. The strict difference follows from a relative entropy identity after the physical catalogue bounds have been proved. When the powers agree, Theorem 2.3 gives the complete full-set spectrum. At and below βh , every positive-area compact initial set has rate γ/β . Above it, one fixed nearly full-area compact set has rate h , whereas the full-set rate increases to $\log 2$. The distinct saturation boundary is $-(\beta/2)\log(a_0 a_1)$. In both cases the compact set is selected before all exponents, precision sequences, and horizons in the respective statement.

Two physical estimates give these results. Theorem 2.5 controls the area covered by any approximately feasible input. Near the origin, a logarithmic slope cone bounds complete feasible intervals without requiring them to map onto the full angular range. The angular product $\lambda_w = \prod a_{w_j}$ and radial product $P_w = \prod r_{w_j}$ enter separately: a first true exit gives a physical area band of size at most $C\delta\lambda_w/P_w$. A finite noncollapsing transient contributes only $O(\delta)$ area. The estimates apply to all feasible prefixes, so no exact strategy is assumed optimal. For the full-set lower bound, Lemma 5.1 constructs actual bounded-run trajectories with a uniform margin against the alternative control. Forcing suitably shorter prefixes gives every frequency contribution to the spectrum before its physical margin falls below the tolerance. An auxiliary radial stopping tree constructs full-horizon upper catalogues, retaining only actual feasible assignments. In the matched case $P_w = \lambda_w^\beta$, and counting all leaf lengths and types evaluates the complete spectrum. Under mismatch, stopping lengths of typical words give the different near-full-area slope. The criterion extends these controlled estimates; it introduces no new pressure or graph-transform method.

Section 7 proves the mismatch bounds and the matching criterion for the same nonlinear first-order family. Section 8 gives both independently perturbed factors with vanishing overlap and an inward boundary example. In the latter, long constant words have empty feasible slices at each fixed positive radius, although every finite word is realised at sufficiently small radius. A constraint-preserving simultaneous conjugacy cannot reduce this example to the displayed boundary-ray-preserving models. Section 9 gives a smooth collapse example explaining why local matching and all-input convergence alone do not suffice.

1.1. Related work. Our finite-horizon count is the outer invariance complexity of Chen and Zhong [1, Section 2.2], with $n+1$ constrained times instead of n . Their fixed-tolerance equivalence results concern bounded catalogue size and finite outer equi-invariability. They do not quantify the catalogue constants along a tolerance sequence linked to the horizon. Stabilization entropies [3, 4] study related information costs under prescribed decay or practical stabilization conditions. Here the task is approximate viability of the actual constraint at every time, rather than a prescribed decay estimate for each state.

Measure-theoretic invariance entropy and variational principles are developed in [10, 9]. Chen, Huang and Zhong [2] study entropy dimensions using specified invariant partition cylinders and time-power weights. Their measures need not be invariant. Those cylinder estimates do not themselves bound the physical sets covered by arbitrary approximate inputs at δ_n . The first-exit estimate and the finite transient in Theorem 2.5 supply this additional geometric connection.

Graph transforms and volume distortion are established methods. Kawan and Da Silva [7, Appendix] give volume estimates near partially hyperbolic all-time lifts. In the present class, every bounded bi-infinite trajectory is the origin: iterating the norm inequality from time $-k$ bounds its time-zero norm by θ^k times a fixed bound. The blow-up angular coordinate is singular there and is not a physical unstable bundle. Their fixed-radius volume estimates

do not provide the joint tolerance estimate needed here. We use the graph-transform method to prove that estimate directly, including truncated feasible images.

Product weights, stopping scales, types and typical sets are classical tools [6, 5]. Yang, Chen, Yang and Zhou [11] establish cost-threshold pressure and Bowen-equation results for a given invariant partition. Such address counts still require a comparison with all approximate physical inputs before they can determine our catalogue. The contribution here is that comparison through actual area and physical witnesses, rather than a new pressure or type formula. The comparisons with [7, 11] use the published graph-transform appendix and the author versions arXiv:1711.01181v2 and arXiv:2309.01628v3, respectively. The estimates used below are proved in full.

2. THE CATALOGUE AND MAIN RESULTS

Let

$$Q = \{(s, z) \in \mathbb{R}^2 : 0 \leq s \leq 1, |z| \leq s\}, \quad \text{area}(Q) = 1.$$

For an infinite binary input $u = (u_1, u_2, \dots)$ set $F_u^0 = \text{id}$ and $F_u^k = F_{u_k} \circ \dots \circ F_{u_1}$. All binary inputs are allowed. Distances are Euclidean, and all logarithms are natural. For a nonempty compact $K \subset Q$ define

$$(2.1) \quad A_u(n, \delta; K) = \{x \in K : \text{dist}(F_u^k x, Q) < \delta \text{ for every } 0 \leq k \leq n\}.$$

Let $r_F(n, \delta; K)$ be the smallest number of infinite inputs whose sets (2.1) cover K . The assignment may depend on the actual initial state. Exit and reentry are allowed subject to every displayed constraint. Only the first n letters affect this count. Controlled feasibility makes it finite.

A precision sequence means $\delta_n > 0$, $\delta_n \rightarrow 0$ and $-\log \delta_n/n \rightarrow \gamma$ for a finite $\gamma \geq 0$. It need not be monotone. Fix

$$(2.2) \quad a_0, a_1 \in (0, 1), \quad a_0 + a_1 = 1, \quad a_0 \neq a_1, \quad \beta_0, \beta_1 > 1,$$

and set $r_i = a_i^{\beta_i}$, $t_i = r_i/a_i < 1$, $a_* = \min_i a_i$, $a^* = \max_i a_i$ and $h = -\sum_i a_i \log a_i$. Write

$$(2.3) \quad A_0 = \begin{pmatrix} r_0 & 0 \\ t_0 a_1 & t_0 \end{pmatrix}, \quad A_1 = \begin{pmatrix} r_1 & 0 \\ -t_1 a_0 & t_1 \end{pmatrix}.$$

The standing assumptions on the actual maps are:

- (1) $F_0, F_1 : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ are C^2 diffeomorphisms, $F_i(0) = 0$ and $DF_i(0) = A_i$.
- (2) $Q \subset F_0^{-1}Q \cup F_1^{-1}Q$.
- (3) For some $\Lambda \geq 1$ and $0 < \theta < 1$, the norm $V(s, z) = \max\{\Lambda|s|, |z|\}$ satisfies

$$(2.4) \quad V(F_i x) \leq \theta V(x) \quad (x \in \mathbb{R}^2, i = 0, 1).$$

The second assumption is one-step controlled invariance; unrestricted concatenation yields an exactly feasible infinite input from each point of Q . The third concerns contraction toward the origin, not distances between two arbitrary trajectories. In particular it controls appended tails even outside Q . Radial motion may depend on angle. No accumulated distortion, word-area bound or catalogue comparison is assumed.

Both matched and mismatched subfamilies are nonempty. For the linear maps A_i , the angular domains $[-1, a_0 - a_1]$ and $[a_0 - a_1, 1]$ cover every fibre. Put $t = \max_i t_i < 1$ and choose $\Lambda = 1 + 4t/(1 - t)$. Then $V(A_i x) \leq \max\{\max_i r_i, t(1 + 1/\Lambda)\}V(x)$ globally, with factor less than one, and $\det A_i = r_i^2/a_i > 0$.

For words write

$$\lambda_w = \prod_{j=1}^{|w|} a_{w_j}, \quad P_w = \prod_{j=1}^{|w|} r_{w_j},$$

with empty products one. Define

$$(2.5) \quad \chi_R = - \sum_i a_i \log r_i > h, \quad \gamma_0 = - \log \max_i r_i > 0, \quad r_0^D + r_1^D = 1.$$

The root D is unique in $(0, 1)$, since the left side decreases from two to zero and $r_0 + r_1 < 1$. Also $\chi_R \geq \gamma_0$.

Theorem 2.1 (First-order matching criterion). *For every fixed system satisfying the standing assumptions, the following property holds if and only if $\beta_0 = \beta_1$: there are $\gamma_1 > 0$ and a function $c : (0, \gamma_1) \rightarrow [0, \infty)$ such that, for every fixed positive-area compact $K \subset Q$, every $\gamma \in (0, \gamma_1)$, and every precision sequence of exponent γ ,*

$$\lim_n n^{-1} \log r_F(n, \delta_n; K) = c(\gamma).$$

Theorem 2.2 (Low-precision mismatch). *Suppose $\beta_0 \neq \beta_1$. For each $\zeta \in (0, 1)$ there is one compact $K_\zeta \subset Q$, of area greater than $1 - \zeta$, such that simultaneously for every $0 < \gamma < \gamma_0$ and every precision sequence of that exponent,*

$$(2.6) \quad \lim_n \frac{\log r_F(n, \delta_n; Q)}{n} = \gamma D, \quad \lim_n \frac{\log r_F(n, \delta_n; K_\zeta)}{n} = \gamma h / \chi_R < \gamma D.$$

The set is selected before the exponent, sequence, and horizon. Every other fixed positive-area compact $K \subset Q$ has lower rate at least $\gamma h / \chi_R$ in this range.

Write H_b for the binary entropy. In the matched case $\beta_0 = \beta_1 = \beta$, the angular cost and spectrum are

$$(2.7) \quad H_b(p) = -p \log p - (1 - p) \log(1 - p), \quad \chi(p) = -p \log a_0 - (1 - p) \log a_1,$$

$$(2.8) \quad S(\gamma) = \max_{p \in [0, 1]} H_b(p) \min\{1, \gamma / [\beta \chi(p)]\},$$

where $0 \log 0 = 0$. In particular $H_b(a_0) = \chi(a_0) = h$.

Theorem 2.3 (Precision spectrum and initial-set dependence). *Suppose $\beta_0 = \beta_1 = \beta$ in the standing assumptions. For every fixed system, every finite $\gamma \geq 0$ and every precision sequence of exponent γ ,*

$$(2.9) \quad \lim_n \frac{\log r_F(n, \delta_n; Q)}{n} = S(\gamma).$$

Every fixed positive-area compact $K \subset Q$ satisfies

$$(2.10) \quad \min\{h, \gamma / \beta\} \leq \liminf_n \frac{\log r_F(n, \delta_n; K)}{n} \leq \limsup_n \frac{\log r_F(n, \delta_n; K)}{n} \leq S(\gamma).$$

Thus the rate exists and is γ / β for $0 \leq \gamma \leq \beta h$. For each $\zeta \in (0, 1)$ there is one fixed compact $K_\zeta \subset Q$ with $\text{area}(K_\zeta) > 1 - \zeta$ such that, simultaneously for every finite $\gamma \geq 0$ and every precision sequence of that exponent,

$$(2.11) \quad \lim_n \frac{\log r_F(n, \delta_n; K_\zeta)}{n} = \min\{h, \gamma / \beta\}.$$

The set is chosen before the exponent, sequence and horizon.

Corollary 2.4 (Closed spectrum and two thresholds). *Under the hypotheses of Theorem 2.3, put $\alpha = a_* \in (0, 1/2)$, $c_b = -\log(1 - \alpha)$ and $d_b = \log((1 - \alpha)/\alpha) > 0$. The common-cost boundary and saturation boundary are respectively*

$$(2.12) \quad \gamma_* = \beta h, \quad \gamma_{\text{sat}} = -\frac{\beta}{2} \log(a_0 a_1) = \beta(c_b + d_b/2) > \gamma_*.$$

The full-set spectrum is

$$(2.13) \quad S(\gamma) = \begin{cases} \gamma/\beta, & 0 \leq \gamma \leq \gamma_*, \\ H_b((\gamma/\beta - c_b)/d_b), & \gamma_* < \gamma < \gamma_{\text{sat}}, \\ \log 2, & \gamma \geq \gamma_{\text{sat}}. \end{cases}$$

The exact gap between Q and the single set K_ζ in the theorem is

$$(2.14) \quad \Delta(\gamma) = S(\gamma) - \min\{h, \gamma/\beta\} = \begin{cases} 0, & 0 \leq \gamma \leq \gamma_*, \\ H_b((\gamma/\beta - c_b)/d_b) - h, & \gamma_* < \gamma < \gamma_{\text{sat}}, \\ \log 2 - h, & \gamma \geq \gamma_{\text{sat}}. \end{cases}$$

It is strictly positive exactly when $\gamma > \gamma_*$. The full-set rate equals $\log 2$ exactly when $\gamma \geq \gamma_{\text{sat}}$.

The first threshold concerns equality of rates across all positive-area compact initial sets; the second concerns the full-set catalogue alone. The theorem does not classify high-side rates of arbitrary compact initial sets or assert that only two such rates are possible.

Theorem 2.5 (Area covered by an arbitrary approximate input). *Under the standing assumptions, whether matched or mismatched, there is a fixed integer $J \geq 1$ with the following property. For every positive-area compact $K \subset Q$ one can choose a compact $T \subset K \setminus \{0\}$ of positive area and $C_J < \infty$ so that for every $e \in (0, h)$ there is $C_e < \infty$ such that*

$$(2.15) \quad \text{area}(T \cap A_u(n, \delta; K)) \leq C_J \delta + C_e \left[e^{-(h-e)m} + \delta e^{(\chi_R - h + 2e)m} \right]$$

for every input u , $n \geq J$, $0 \leq m \leq n - J$ and $0 < \delta \leq 1$. The integer and the set precede e, n, δ, u . For $K = Q$ the set can have area greater than $1 - \zeta$ for any prescribed $\zeta \in (0, 1)$.

3. LOCAL FEASIBLE GEOMETRY

Use (s, y) with $z = sy$, only for $s > 0$. Define

$$R_i(s, y) = F_i^s(s, sy)/s, \quad N_i(s, y) = F_i^z(s, sy)/s, \quad T_i = N_i/R_i.$$

The integral formula $F_i(s, sy)/s = \int_0^1 DF_i(ts, tsy)(1, y) dt$ extends R_i, N_i as C^1 functions to $s = 0$. At zero

$$(3.1) \quad R_i(0, y) = r_i, \quad L_0(y) := T_0(0, y) = (y + a_1)/a_0, \quad L_1(y) := T_1(0, y) = (y - a_0)/a_1.$$

We choose the local constants from the maps as follows. Let L bound the Euclidean bilinear operator norm of both second derivatives on $\overline{B}(0, 2)$, and let $r_* = \min r_i$, $r^* = \max r_i$, $a^* = \max a_i$, $a_* = \min a_i$. Take $B = 10^4(1 + L)/r_*^3$, and shrink $\rho > 0$ so that

$$(3.2) \quad \rho \leq \min \left\{ \frac{1}{4}, \frac{r_*}{8(1 + L)}, \frac{1 - r^*}{8B}, \frac{1/a^* - 1}{16B}, \frac{1}{16B}, \frac{a_*}{16B} \right\}.$$

Taylor's integral formula gives $|R_i - r_i| \leq 2Ls$, $|\partial_s R_i|, |\partial_s N_i| \leq 2L$, and $|\partial_y R_i| \leq 2Ls$, $|\partial_y N_i - t_i| \leq 2Ls$. The same bounds apply with the appropriate constant linear term in N_i . In particular $R_i \geq r_*/2$. Dividing these formulas by R_i , and differentiating the quotient, gives, with the generous common constant B ,

$$(3.3) \quad \begin{aligned} |R_i - r_i| &\leq Bs, & |\log(R_i/r_i)| &\leq Bs, \\ |\partial_s \log R_i| &\leq B, & |\partial_y \log R_i| &\leq Bs, \\ |T_i - L_i| &\leq Bs, & |\partial_s T_i| &\leq B, & |\partial_y T_i - 1/a_i| &\leq Bs. \end{aligned}$$

For example $|N_i| \leq 3$ on this rectangle; quotient derivatives use $R_i^{-2} \leq 4/r_*^2$. Thus the displayed choice of B dominates all the constants in (3.3). These are derived local inequalities, not extra hypotheses.

Set $\bar{r} = r^* + B\rho < 1$ and $\hat{r} = r^* + 2B\rho < 1$. On an exactly feasible local orbit, $s_k \leq \bar{r}^k s_0$. The bounds in (3.2) also give $T_1(s, -1) < -1$ and $T_0(s, 1) > 1$. Controlled invariance therefore forces

$$(3.4) \quad T_0(s, -1) \geq -1, \quad T_1(s, 1) \leq 1.$$

Equality is not required in (3.4). The increasing functions $T_i(s, \cdot)$ have unique cut points $\vartheta_0(s)$ with $T_0 = 1$ and $\vartheta_1(s)$ with $T_1 = -1$. They equal $a_0 - a_1 + O(s)$. Their feasible intervals are respectively $[-1, \vartheta_0(s)]$ and $[\vartheta_1(s), 1]$. Controlled invariance gives $\vartheta_1 \leq \vartheta_0$. Thus any local overlap has angular width $O(s)$ and physical width $O(s^2)$. Positivity of the overlap is not assumed.

3.1. Complete intervals with truncated images. Consider a radial graph $s = S(y)$, on any closed subinterval of $[-1, 1]$, with $|S'/S| \leq 1$. Put $k = S'/S$. For either input its terminal angular derivative along the graph is

$$J_i = \partial_y T_i + \partial_s T_i S' = 1/a_i + O(BS) > 0.$$

The new radial graph has logarithmic slope

$$(3.5) \quad k_{\text{new}} = \frac{k + (\partial_s \log R_i)S' + \partial_y \log R_i}{J_i}.$$

Its numerator is at most $1 + 2B\rho$ in absolute value and its denominator at least $1/a^* - 2B\rho$. By (3.2), $|k_{\text{new}}| \leq 1$. Piecewise C^1 graphs obey the same assertion almost everywhere.

For a fixed initial radius $0 < s_0 \leq \rho$, start with the horizontal graph $S \equiv s_0$. Let $I_w(s_0)$ be the set of initial angles whose trajectory under a finite word w stays in Q at every intermediate time. It is a closed interval, possibly empty or a point. Here is the induction that checks all constraints. The current angular image is an interval, with a graph obeying (3.5). Extend that graph constantly to the two sides of its domain. The extension is positive, bounded by ρ , and obeys the logarithmic-slope bound almost everywhere. Both $T_i(S(y), y)$ are increasing. By (3.4), control 0 cannot violate the lower angular boundary, and control 1 cannot violate the upper one. The other boundary imposes one cut. Pulling that cut back through the increasing preceding angular map leaves an interval. No terminal map is asserted to be onto $[-1, 1]$. The two allowed children cover the parent, by controlled invariance, but either child can be empty.

Along any such feasible graph, (3.3) and the geometric decay of s_j give constants independent of w, s_0 and its initial angle:

$$(3.6) \quad C_R^{-1} s_0 P_w \leq s_{|w|} \leq C_R s_0 P_w, \quad C_A^{-1} \lambda_w^{-1} \leq \frac{dY_w}{dy_0} \leq C_A \lambda_w^{-1}.$$

For instance one may take $C_R = \exp(B\rho/(1 - \bar{r}))$, $C_A = \exp(4B\rho/(1 - \bar{r}))$: $|a_i J_i - 1| \leq 2Bs_j \leq 1/8$, so its logarithm has modulus at most $4Bs_j$. The first estimate sums $\log(R_i/r_i)$; the second sums the angular derivatives. They are independent product estimates. Only in the matched case is $P_w = \lambda_w^\beta$. Integration over the *actual* angular image implies

$$(3.7) \quad |I_w(s_0)| \leq 2C_A \lambda_w, \quad |\mathcal{I}_w| \leq C_A \rho^2 \lambda_w, \quad \mathcal{I}_w = \{(s, sy) : 0 < s \leq \rho, y \in I_w(s)\}.$$

The Jacobian of $(s, y) \mapsto (s, sy)$ is s . There is no positive lower bound on $|I_w|$; some fixed-radius slices in Section 8.2 are empty.

For a point interval the length bound is zero. Derivative comparisons on nondegenerate intervals extend to their endpoints by continuity; degenerate fibres remain in all covers and do not contribute area.

3.2. The first true exit. Suppose the prefix w of length $j - 1$ is exactly feasible and the next prescribed input i is the first one to leave Q . On the current graph the map T_i is increasing with derivative at least $1/(2a_i)$. The radial output is at least cs_0P_w , by (3.3), (3.6), with $c > 0$ fixed. If this wrong step nonetheless has distance less than δ from Q , then

$$(3.8) \quad (|z_j| - s_j)_+ < 2\delta.$$

This follows from the Euclidean Lipschitz linear functionals $z - s$ and $-z - s$. The output radius is positive and at most ρ , so the violated constraint is angular, not the upper radius. Consequently the wrong part of the terminal angular interval has diameter at most $C\delta/(s_0P_w)$. By (3.6), its initial-angle preimage has length at most

$$C\delta\lambda_w/(s_0P_w).$$

After integration against $s_0 ds_0$, its physical area is at most

$$(3.9) \quad C\delta\lambda_w/P_w.$$

This argument also covers a point interval or a terminal interval lying wholly beyond the cut: monotonicity bounds the diameter of the part satisfying (3.8), without requiring a cut point inside the actual image. It does not require the wrong trajectory to remain outside afterwards. Any later reentry cannot erase constraint (3.8) at its first exit. A different exact feasible coding is not counted as an exit and has no role in (3.9).

4. ALL FEASIBLE PREFIXES AND ACTUAL COVERED AREA

Proof of Theorem 2.5. For any $d > 0$, the total weight $\sum \lambda_w$ of length- m words with $|\#0/m - a_0| > d$ is at most $2e^{-2d^2m}$. To see this without importing a new result, regard the weights as Bernoulli probabilities. For a Bernoulli variable X , $\log \mathbb{E}e^{t(X-a_0)} \leq t^2/8$, since the second derivative of this logarithm is a tilted Bernoulli variance, at most $1/4$; apply the exponential Markov inequality with $t = 4d$ and with $-4d$. By (3.7), the area of the union of the corresponding complete feasible regions is summable in m . Overlaps do not invalidate the union bound. Borel–Cantelli for rational $d > 0$ shows that for area-almost every $x \in Q_\rho$, all feasible length- m prefixes at x have zero frequency tending to a_0 , uniformly over those prefixes. Here $Q_\rho = Q \cap \{s \leq \rho\}$. Controlled invariance guarantees at least one such prefix for every m ; the origin can be discarded as an area-zero point.

Choose $J \geq 1$ so large that $\theta^J \sup_Q V \leq \rho$. Every exactly feasible length- J prefix then ends in Q_ρ . For each of the finitely many such words σ , let $D_\sigma \subset Q$ be its complete feasible initial region. Each F_σ is a diffeomorphism, so the pullback of the local exceptional null set is null. Hence on almost every initial state of Q , all exact feasible transient prefixes and all their feasible length- m tails have frequencies tending uniformly to a_0 . The maximum over the finitely many transient choices has this same property. These maxima are measurable: they are maxima over finitely many closed finite-word feasibility regions.

Egorov followed by compact inner approximation gives a compact $T \subset K \setminus \{0\}$ of positive area on which this convergence is uniform. For $K = Q$ it gives $|T| > 1 - \zeta$. This single set is chosen using frequency convergence, without choosing a precision exponent or sequence. For every $e > 0$ there is $D_e \geq 1$ such that every feasible tail w , after every feasible σ from every state of T , obeys

$$(4.1) \quad \begin{aligned} D_e^{-1}e^{-(h+e)|w|} &\leq \lambda_w \leq D_e e^{-(h-e)|w|}, \\ D_e^{-1}e^{-(\chi_R+e)|w|} &\leq P_w \leq D_e e^{-(\chi_R-e)|w|}. \end{aligned}$$

Both logarithmic costs are affine functions of word frequency. Finitely many short lengths are absorbed in D_e , on the same set T for every e .

The finite transient does not silently assume exact feasibility for an approximate input. The polygon Q has $|Q_\delta \setminus Q| \leq C\delta$ for $0 < \delta \leq 1$. For all prefixes of length at most J ,

the inverse Jacobian determinants are bounded on the compact $\overline{Q_1}$. If an approximately feasible input first exits during these J steps, its initial state belongs to a pullback of this boundary tube. Summing over its at most J possible first-exit times, with the maximum over the finite prefix family, gives

$$(4.2) \quad |\{x \in Q : \text{this input first exits by time } J \text{ and } x \in A_u(n, \delta; K)\}| \leq C_J \delta.$$

No bounded-distortion assumption is hidden here: the determinant bound exists because these are finitely many actual diffeomorphisms. It is uniform in the continuation, horizon, and input.

Fix an arbitrary input and split its covered states into (4.2) and the states exact through time J . On the latter, let σ be its actual length- J prefix and use F_σ to transport the states to Q_ρ . Its inverse area distortion is a fixed constant. Exact feasibility through the next m steps has area at most $C\lambda_{u_{J+1}\dots u_{J+m}}$, bounded using (4.1). For a first true exit at tail time $j \leq m$, its length- $j-1$ prefix intersects the transported set, so (4.1) applies to that prefix and (3.9) bounds its area. The ratio of the two products in (4.1) is at most $D_e^2 e^{(\chi_R - h + 2e)(j-1)}$. Since $\chi_R > h$, summing the first-exit classes gives at most $C_e \delta e^{(\chi_R - h + 2e)m}$. This proves (2.15), including all inputs and all later reentries. $\square$

5. PHYSICAL TRAJECTORIES WITH A WRONG-CONTROL MARGIN

Lemma 5.1 (Bounded-run realisation). *For each integer $L \geq 2$, there is a positive radius s_L and $d'_L > 0$ such that every infinite binary sequence whose successive runs have length at most L is realised by an actual exactly feasible orbit starting at $(s_L, s_L y_0)$. At every step of that orbit the alternative control is genuinely infeasible, with an angular excess at least d'_L . The radius and margin are uniform over the sequences. No such assertion is made for unbounded runs or all constant words.*

Proof. The inverse reference maps are $f_0(y) = a_0 y - a_1$, $f_1(y) = a_1 y + a_0$. For any fixed infinite input v , successive inverse compositions give a unique reference bounded orbit $y_k^0 \in [-1, 1]$. If runs have length at most L , the next opposite digit is seen within $L + 1$ digits. Since $f_0(y) + 1 = a_0(y + 1)$, $1 - f_1(y) = a_1(1 - y)$, and each opposite digit inserts an endpoint distance at least $2a_*$,

$$(5.1) \quad |y_k^0| \leq 1 - d_L, \quad d_L = 2a_*^{L+1} > 0.$$

Let $b = d_L/4$ and consider the complete closed ball of sequences $\|y - y^0\|_\infty \leq b$ in ℓ^∞ . For each such sequence and a positive fixed initial radius s_* , recursively define $s_{k+1} = s_k R_{v_{k+1}}(s_k, y_k)$. These radii stay below $\bar{r}^k s_*$, independently of whether the angular recursion has yet been imposed. The scalar map $s \mapsto s R_i(s, y)$ has derivative bounded in absolute value by $\hat{r} < 1$, and its y derivative is bounded by Bs^2 , since $R_i \leq \bar{r} < 1$ and (3.3) bounds its logarithmic angular derivative. Thus two candidate sequences give

$$(5.2) \quad \sup_k |s_k(y) - s_k(\tilde{y})| \leq C_s s_*^2 \|y - \tilde{y}\|_\infty, \quad C_s = B/(1 - \hat{r}).$$

Write $e_i(s, y) = T_i(s, y) - L_i(y)$. By (3.3), $|e_i| \leq Bs$, $|\partial_s e_i| \leq B$, and $|\partial_y e_i| \leq Bs$. Define

$$(5.3) \quad (\mathcal{B}y)_k = f_{v_{k+1}}(y_{k+1}) - a_{v_{k+1}} e_{v_{k+1}}(s_k(y), y_k).$$

Choose $s_* \leq \rho$ so small that

$$(5.4) \quad a^* B s_* \leq (1 - a^*)b/2, \quad a^* B(s_* + C_s s_*^2) < (1 - a^*)/2.$$

Then (5.3) maps the ball into itself and has Lipschitz constant at most $a^* + a^* B(s_* + C_s s_*^2) < 1$. The contraction mapping theorem gives a fixed sequence. Its equation is exactly $y_{k+1} = T_{v_{k+1}}(s_k, y_k)$, not a formal symbolic substitution. It has $|y_k| \leq 1 - 3d_L/4$, so the constructed physical trajectory is exactly feasible for all times.

To check the alternative control, suppose the chosen digit is zero. Then $T_0(s_k, y_k) = y_{k+1} \leq 1 - 3d_L/4$, and (3.3) gives

$$y_k \leq a_0 - a_1 - 3a_0d_L/4 + a_0Bs_k.$$

Substitution into $T_1 = L_1 + e_1$ yields

$$T_1(s_k, y_k) \leq -1 - \frac{3a_0d_L}{4a_1} + B(1 + a_0/a_1)s_k.$$

For chosen digit one, the symmetric computation gives

$$T_0(s_k, y_k) \geq 1 + \frac{3a_1d_L}{4a_0} - B(1 + a_1/a_0)s_k.$$

Shrink s_* once more, before considering any particular sequence, so that these last error terms are at most $a_*d_L/(4a_*)$. Both wrong angular excesses are then at least $d'_L = a_*d_L/(2a_*)$. Set $s_L = s_*$. All choices depend only on L and the fixed system. This proves the lemma, including the physical realisation and the genuine-infeasibility assertion. $\square$

Lemma 5.2 (Forcing a block prefix). *Fix $L \geq 2$ and $1 \leq j \leq L - 1$, and put*

$$M_L = \binom{L-2}{j-1}, \quad c_L = -\frac{j}{L} \log r_0 - \left(1 - \frac{j}{L}\right) \log r_1.$$

For every precision sequence of exponent $\gamma > 0$ and every $0 < t < \min\{1, \gamma/c_L\}$, for all sufficiently large n ,

$$(5.5) \quad r_F(n, \delta_n; Q) \geq M_L^{\lfloor tn/L \rfloor}.$$

Proof. Use all length- L blocks starting with zero, ending with one, and containing j zeros. Every concatenation has runs of length at most L and radial product e^{-Lc_L} per complete block. Put $k_n = \lfloor tn/L \rfloor$ and $m_n = Lk_n$. For each of the $M_L^{k_n}$ initial concatenations, append one fixed allowable block forever. Lemma 5.1 realises these infinite inputs at the same positive radius s_L . Every prefix through time m_n has $P_{v|k} \geq P_{v|m_n} = e^{-c_L m_n}$.

If an arbitrary catalogue input first differs from a witness at $j' \leq m_n$, its preceding states agree with that witness. By (3.6), their radius at $j' - 1$ is at least $C_R^{-1} s_L e^{-c_L m_n}$. The wrong radial factor is at least $r_*/2$, and the alternative angular excess is at least d'_L . Its actual physical excess therefore obeys

$$(5.6) \quad |z_{j'}| - s_{j'} \geq E_L e^{-c_L m_n}, \quad E_L = \frac{s_L r_* d'_L}{2C_R} > 0.$$

The ratio to $2\delta_n$ has logarithmic rate $\gamma - c_L t > 0$. For large n , it violates (3.8), regardless of later reentry. Thus the approximate input must agree with all m_n witness digits. Distinct selected prefixes give distinct actual states: at their first different digit one input is feasible and the other has the proved wrong-control margin. One catalogue input covers at most one of them, proving (5.5). The block length, positive radius, and margin remain fixed throughout this horizon limit. Integer rounding and every $o(n)$ precision fluctuation are absorbed by the strict choice of t . $\square$

6. STOPPING CATALOGUES AND THE MATCHED SPECTRUM

We first give a catalogue construction under the standing assumptions. The reference tree provides an upper bound; it is not presumed necessary or optimal.

Lemma 6.1 (A complete radial stopping catalogue). *Put $M = \max\{1, 4\sqrt{2}\Lambda C_R \rho\}$. For $N \geq 0$ and $0 < t \leq 1$, let $\mathcal{L}_P(N, t)$ be the full reference binary tree stopped at the first $P_w \leq t$, or at depth N , whichever occurs first. For $n \geq J$ and $0 < \delta \leq 1$,*

$$(6.1) \quad r_F(n, \delta; Q) \leq 2^J \# \mathcal{L}_P(n - J, \delta/M).$$

The catalogue can retain only transient/leaf pairs whose complete actual feasible initial region is nonempty.

Proof. For every actual initial state choose an exactly feasible infinite input. Its first J letters form a feasible transient ending in Q_ρ ; the subsequent letters reach a reference leaf. Keep every pair realised by an actual state, including singleton regions and boundary states. Before stopping the assigned states are exactly feasible. At a threshold leaf, (3.6) gives

$$s_{|w|} \leq C_R \rho P_w \leq C_R \rho \delta / M \leq \delta / (4\sqrt{2}\Lambda).$$

Since $V = \Lambda s$ on Q , appending the same infinite zero tail and using (2.4) keeps the Euclidean norm at most $\delta/4$ at every later time. The origin belongs to Q , so every constraint holds strictly on that tail. A horizon leaf is exactly feasible through time n , and its later extension is irrelevant. Every constructed input fixes the origin as well. There are at most 2^J transients, proving (6.1). $\square$

6.1. Upper bound for the matched spectrum. For the rest of this section assume $\beta_0 = \beta_1 = \beta$. Then $P_w = \lambda_w^\beta$, $\chi_R = \beta h$, and

$$t_\delta = (\delta/M)^{1/\beta}, \quad b_\delta = -\log t_\delta.$$

Lemma 6.1 uses the tree stopped at $\lambda_w \leq t_\delta$ or at depth $N = n - J$. Put $c_i = -\log a_i$, $c_{\min} = \min_i c_i > 0$, $c_{\max} = \max_i c_i$, and $\Gamma_{n,\delta} = \beta(b_\delta + c_{\max})/n$. A threshold leaf has logarithmic angular cost in $[b_\delta, b_\delta + c_{\max}]$; an unfinished horizon leaf has cost below b_δ . Thus every nonempty leaf of length $\ell \leq n$, containing j zeros with $p = j/\ell$, satisfies

$$(6.2) \quad \ell \chi(p) < b_\delta + c_{\max}, \quad \frac{\ell}{n} \leq \min\{1, \Gamma_{n,\delta}/[\beta \chi(p)]\}.$$

The corresponding term of the binomial law of parameter j/ℓ gives $\binom{\ell}{j} \leq e^{\ell H_b(j/\ell)}$. Endpoint types have one word. Each leaf type therefore has at most $e^{nS(\Gamma_{n,\delta})}$ words. There are at most $(n+1)^2$ length/type pairs, including a possible empty word. By the actual catalogue construction,

$$(6.3) \quad r_F(n, \delta; Q) \leq 2^J (n+1)^2 e^{nS(\Gamma_{n,\delta})}.$$

No nonempty fixed-radius slice or full terminal image is assumed.

Every function under the maximum defining S is Lipschitz in γ , with common constant at most $\log 2/(\beta c_{\min})$. Their maximum is continuous. Here $\beta b_{\delta_n} = -\log \delta_n + \log M$, so $\Gamma_{n,\delta_n} \rightarrow \gamma$ for every precision sequence. The fixed transient and polynomial factors have zero rate, giving

$$(6.4) \quad \limsup_n n^{-1} \log r_F(n, \delta_n; Q) \leq S(\gamma).$$

This includes $\gamma = 0$ and nonmonotone sequences.

The reference angular leaf weights sum to one, and each is at least $a_* t_\delta$: a threshold leaf has parent weight above t_δ , whereas an unfinished leaf has weight above t_δ itself. This identity concerns a reference tree, not a physical partition. Enlarging a constant for $n < J$ yields the useful coarse bound

$$(6.5) \quad r_F(n, \delta; Q) \leq C_U \delta^{-1/\beta} \quad (0 < \delta \leq 1), \quad r_F(n, \delta; Q) \leq 2^n.$$

6.2. Shorter physical prefixes give the lower bound. Let $\gamma > 0$ and fix $p \in (0, 1)$. Choose $j_L \in \{1, \dots, L-1\}$ with $j_L/L \rightarrow p$, and put

$$M_L = \binom{L-2}{j_L-1}, \quad B_L = L^{-1} \log M_L, \quad \chi_L = \chi(j_L/L).$$

The exact identity and elementary type bounds

$$\binom{L-2}{j-1} = \frac{j(L-j)}{L(L-1)} \binom{L}{j}, \quad \frac{e^{LH_b(j/L)}}{L+1} \leq \binom{L}{j} \leq e^{LH_b(j/L)}$$

give $B_L \rightarrow H_b(p)$ and $\chi_L \rightarrow \chi(p)$. For the lower type bound the j -th term is a largest term of the binomial law with parameter j/L , so is at least $1/(L+1)$. The additional fraction has logarithm $o(L)$ for interior p .

The radial block cost in Lemma 5.2 is now $c_L = \beta\chi_L$. Fix L first and take

$$(6.6) \quad 0 < t < \min\{1, \gamma/(\beta\chi_L)\}.$$

The lemma, which checks the physical margin and every constrained time, gives

$$(6.7) \quad \liminf_n n^{-1} \log r_F(n, \delta_n; Q) \geq tB_L.$$

For fixed L take the supremum over the strict t , next let $L \rightarrow \infty$, and finally take the supremum over $p \in (0, 1)$. Endpoint frequencies have zero entropy, so continuity yields $\liminf n^{-1} \log r_F(n, \delta_n; Q) \geq S(\gamma)$. No witness radius or margin is required to remain uniformly positive as $L \rightarrow \infty$; they are fixed in each horizon limit. For $\gamma = 0$, the bound $r_F \geq 1$ and (6.4) give the zero limit. This proves (2.9).

6.3. Positive-area and fixed near-full initial sets. To complete Theorem 2.3, apply Theorem 2.5 to a fixed positive-area compact K and its positive-area set T . A catalogue covering K has at least $\text{area}(T)/\sup_u \text{area}(T \cap A_u(n, \delta_n; K))$ elements. For fixed $e \in (0, h)$, put

$$m_n = \min \left\{ n - J, \left\lfloor \frac{-\log \delta_n}{\chi_R + e} \right\rfloor \right\}.$$

Eventually $m_n \geq 0$, and $\delta_n \leq e^{-(\chi_R + e)m_n}$. Each term on the right of (2.15) is then at most a constant times $e^{-(h-e)m_n}$. Consequently

$$\liminf_n n^{-1} \log r_F(n, \delta_n; K) \geq (h - e) \min\{1, \gamma/(\chi_R + e)\}.$$

Since $\chi_R = \beta h$ here, letting $e \downarrow 0$ gives the lower bound in (2.10); for $\gamma = 0$ the bound $r_F \geq 1$ also suffices. The upper bound follows from $r_F(n, \delta; K) \leq r_F(n, \delta; Q)$ and (2.9). The evaluation of S below will give the claimed common low-side rate.

For $\zeta \in (0, 1)$ take the single near-full set T constructed in Section 4 and put $K_\zeta = T \cup \{0\}$. All exactly feasible length- n words meeting T have, after their J -digit transient, frequency uniformly tending to a_0 . If $d_N \rightarrow 0$ bounds this deviation for tails of length $N = n - J$, there are at most

$$2^J(N+1) \exp \left(N \max_{|p-a_0| \leq d_N} H_b(p) \right) = e^{(h+o(1))n}$$

such words. Every state of T has an exactly feasible prefix, so these words, extended by any infinite tail, cover T through time n . One additional input covers the origin. This gives an upper rate h for every precision sequence, without selecting a strategy. Bound (6.5) also gives upper rate γ/β . The lower bound already proved gives (2.11). The construction of T used uniform frequency convergence and compact inner approximation only; it preceded every exponent, precision sequence and horizon. This completes Theorem 2.3 subject to the following elementary evaluation.

6.4. Evaluation of the maximum.

Proof of Corollary 2.4. Reparameterise the maximum by the frequency q of the smaller branch $\alpha = a_*$. This only changes the optimisation variable, not the dynamics or control assignments. Then $\chi = c_b + d_b q$. Write $x = \gamma/\beta$. The entropy inequality

$$c_b + d_b q - H_b(q) = q \log(q/\alpha) + (1 - q) \log((1 - q)/(1 - \alpha)) \geq 0$$

follows from $-\log v \geq 1 - v$ and extends to the endpoints by continuity; equality holds at $q = \alpha$. For $x \leq h$, the objective $H_b(q) \min\{1, x/(c_b + d_b q)\}$ is at most x and equals x at α .

For $h < x < c_b + d_b/2$ put $q_x = (x - c_b)/d_b \in (\alpha, 1/2)$. When $q \leq q_x$ the objective is $H_b(q) \leq H_b(q_x)$. When $q \geq q_x$ it is $xH_b(q)/(c_b + d_b q)$. This ratio decreases for $q > \alpha$: the numerator of its derivative is

$$D(q) = (c_b + d_b q)H'_b(q) - d_b H_b(q), \quad D(\alpha) = 0, \quad D'(q) = (c_b + d_b q)H''_b(q) < 0.$$

The maximum is therefore $H_b(q_x)$. Finally, for $x \geq c_b + d_b/2$, the value at $q = 1/2$ is $\log 2$, the universal entropy maximum. Since $\alpha < 1/2$, $c_b + d_b/2 > c_b + d_b \alpha = h$. The three pieces match at the thresholds, proving (2.12)–(2.13) and the low-side assertion of Theorem 2.3. Subtracting (2.11) gives (2.14). Strict increase of H_b on $(\alpha, 1/2)$ proves the stated strict gap and the exact saturation threshold. $\square$

Corollary 6.2 (Matched linear systems). *If $\beta_0 = \beta_1 = \beta$, the pair (A_0, A_1) has all the conclusions of Theorems 2.3 and 2.5.*

Proof. The angular maps are $(y + a_1)/a_0$ and $(y - a_0)/a_1$. Their feasible domains $[-1, a_0 - a_1]$ and $[a_0 - a_1, 1]$ cover the fibre. Put $t = \max_i t_i < 1$, choose $\Lambda = 1 + 4t/(1 - t)$ and set $v = t(1 + 1/\Lambda) < 1$. Then $V(A_i x) \leq \max\{\max_i r_i, v\}V(x)$ globally. The remaining hypotheses are immediate. $\square$

7. MISMATCH AND THE COMMON-COST CRITERION

We now return to the standing assumptions with $\beta_0 \neq \beta_1$. The same physical estimates apply, but the angular and radial products no longer have a common-power identity.

Proof of Theorem 2.2. First consider the upper rate on Q . For every $0 < \gamma < \gamma_0$, all branches of the tree in Lemma 6.1 cross $P_w \leq \delta_n/M$ before depth $n - J$ for large n , since

$$P_w \leq e^{-\gamma_0(n-J)} < \delta_n/M \quad \text{at that depth.}$$

The ratio has logarithmic rate $\gamma - \gamma_0 < 0$, so the statement includes every $o(n)$ tolerance fluctuation. For the uncapped radial tree, write $N(t)$ for its leaf count. Replacing a parent by its children preserves $\sum P_w^D = 1$, and each leaf satisfies $r_* t < P_w \leq t$. Hence

$$(7.1) \quad t^{-D} \leq N(t) \leq (r_* t)^{-D}, \quad r_F(n, \delta_n; Q) \leq 2^J N(\delta_n/M).$$

This proves the upper rate γD through an actual full-time catalogue, without asserting that the reference tree is necessary.

For the lower rate put $p_* = r_0^D$, so $1 - p_* = r_1^D$, and $c_R(p) = -p \log r_0 - (1 - p) \log r_1$. Then

$$(7.2) \quad H_b(p_*) = D c_R(p_*).$$

Choose $j_L/L \rightarrow p_*$ and use the blocks in Lemma 5.2, with $c_L = c_R(j_L/L)$ and $B_L = L^{-1} \log \binom{L-2}{j_L-1}$. The binomial calculation in Section 6 gives $B_L \rightarrow H_b(p_*)$ and $c_L \rightarrow c_R(p_*)$. For each fixed L , $c_L \geq \gamma_0 > \gamma$. Every strict $0 < t < \gamma/c_L < 1$ therefore gives lower rate tB_L . Take its supremum first, then let $L \rightarrow \infty$. Equation (7.2) gives the lower rate γD . The positive-radius witnesses and wrong-control margins are those of the common physical lemma, not midpoints of assumed full images. This proves the first limit in (2.6).

For any fixed positive-area compact K , take the set T in Theorem 2.5, and set

$$m_n = \min \left\{ n - J, \left\lfloor \frac{-\log \delta_n}{\chi_R + e} \right\rfloor \right\}.$$

For large n it is nonnegative, and $\delta_n \leq e^{-(\chi_R + e)m_n}$. Every term of (2.15) is then bounded by a constant times $e^{-(h-e)m_n}$. Covering the positive area of T implies lower rate $\gamma(h - e)/(\chi_R + e)$, since $\gamma < \gamma_0 \leq \chi_R$. Let $e \downarrow 0$. This proves the asserted universal lower bound $\gamma h/\chi_R$.

For $K = Q$, choose the single set T of area greater than $1 - \zeta$ in the area proof, and put $K_\zeta = T \cup \{0\}$. Keep only stopped pairs in Lemma 6.1 realised on T . They cover it with the complete tails already constructed. Let $q_\delta = -\log(\delta/M)$ and $c_{\max} = -\log r_*$. Each retained tail of length ℓ has

$$q_\delta \leq -\log P_w < q_\delta + c_{\max}, \quad \ell \leq \frac{q_\delta + c_{\max} + \log D_e}{\chi_R - e},$$

by (4.1). At this length, $\lambda_w \geq D_e^{-1} e^{-(h+e)\ell}$, while all reference length- ℓ weights sum to one. There are at most $D_e e^{(h+e)\ell}$ such words. Sum over these lengths and the at most 2^J transients, and include one input for the origin. For fixed $e \in (0, h)$ this gives

$$(7.3) \quad \log r_F(n, \delta_n; K_\zeta) \leq \frac{h+e}{\chi_R - e} [q_{\delta_n} + c_{\max} + \log D_e] + O_e(\log(1 + q_{\delta_n})) + O_e(1).$$

Because $q_{\delta_n} = \gamma n + o(n)$, letting $e \downarrow 0$ gives upper rate $\gamma h / \chi_R$, matching the lower bound. The set was selected through uniform frequency convergence before e , the precision exponent, the sequence, and the horizon.

Finally let $b_i = r_i^D$. The classical relative-entropy identity is

$$D\chi_R - h = \sum_i a_i \log(a_i/b_i) \geq 0.$$

Concavity of the logarithm proves the inequality, with equality only when $b_i = a_i$ for both labels. Equality would require $\beta_0 D = \beta_1 D = 1$, contrary to mismatch. Thus $D > h/\chi_R$, completing the theorem. $\square$

Proof of Theorem 2.1. If $\beta_0 = \beta_1 = \beta$, Theorem 2.3 gives the common rate $c(\gamma) = \gamma/\beta$ on $(0, \beta h)$. If $\beta_0 \neq \beta_1$, Theorem 2.2 separates Q and one fixed nearly full-area compact set for every $0 < \gamma < \gamma_0$. Any interval adjacent to zero contains such an exponent, contradicting the common-rate property. $\square$

The criterion concerns precisely the first derivatives (2.3), global invertibility, controlled feasibility of Q , and the common norm inequality. The proof does not extend this necessity statement to arbitrary feedback systems, classify high-precision mismatch rates, or compare a fixed strategy with the optimal catalogue.

8. APPLICATIONS OF THE CRITERION

The next two subfamilies use $\beta_0 = \beta_1 = \beta$ and illustrate the matched theorem beyond exact linear or boundary-ray geometry.

8.1. Independent factors and vanishing overlap. This subfamily shows how the criterion applies to separately varying branch lengths and radial factors. Fix a smooth function $0 \leq \kappa \leq 1$ equal to one on $\overline{B}(0, 3/2)$ and with support contained in $B(0, 7/4)$. For example, with $E(v) = e^{-1/v}$ for $v > 0$ and $E(v) = 0$ otherwise, take

$$\kappa(x) = \frac{E(169/64 - |x|^2)}{E(169/64 - |x|^2) + E(|x|^2 - 9/4)}.$$

Its denominator is positive and its support is the closed ball of radius $13/8$. Set

$$C_\kappa = \max\{1, \max_{|\alpha| \leq 2} \|\partial^\alpha(s\kappa)\|_\infty\}, \quad C = 1 + C_\kappa, \quad d = -\log a^*,$$

$$\nu_0 = \frac{1}{C} \min \left\{ \frac{a_*}{1000}, \frac{(\beta - 1)d}{4(1 + 2/a_*)} \right\} > 0.$$

For $0 < \nu < \nu_0$, take $\eta, \xi_0, \xi_1 \in C_c^\infty(B(0, 2))$, zero at the origin, with all partial derivatives through order two bounded by ν . Take $0 \leq u < \nu$ and define at the input state

$$P_0 = a_0 + \eta + us\kappa, \quad P_1 = a_1 - \eta + us\kappa, \quad \mathcal{R}_i = a_i^\beta e^{\xi_i}.$$

Consider

$$(8.1) \quad \begin{aligned} \tilde{F}_0(s, z) &= (s\mathcal{R}_0, (\mathcal{R}_0/P_0)[z + (1 - P_0)s]), \\ \tilde{F}_1(s, z) &= (s\mathcal{R}_1, (\mathcal{R}_1/P_1)[z - (1 - P_1)s]). \end{aligned}$$

The transverse coefficient is $1 - P_i$. No identity $\mathcal{R}_i = P_i^\beta$ is imposed.

Corollary 8.1. *Every pair (8.1) satisfies the hypotheses and conclusions of Theorems 2.3 and 2.5. For $u > 0$, both inputs are feasible on a set with nonempty interior, whose fibre width near the origin is of order us^2 .*

Proof. Put $\mu = C\nu$, $p_- = a_* - \mu$, $p_+ = 1 - p_-$, $r = e^\mu(a^*)^\beta$ and $t = e^\mu(a^*)^{\beta-1}/(1 - \mu/a_*)$. All perturbations $P_i - a_i$ and ξ_i , including their first two derivatives, are bounded by μ . They vanish at the origin. Globally $p_- \leq P_i \leq p_+$, $\mathcal{R}_i \leq r$ and $\mathcal{R}_i/P_i \leq t$. Since $-\log(1 - \mu/a_*) \leq 2\mu/a_*$,

$$\log t \leq -(\beta - 1)d + (1 + 2/a_*)\mu < 0, \quad \log r < 0.$$

For $\Lambda = (1 + t)/(1 - t)$, the transverse coefficients on z and s are at most t , giving the global norm factor $\max\{r, 2t/(1 + t)\} < 1$.

For invertibility use $J_i(s, z) = (s, z - \sigma_i s) = (s, v)$, where $\sigma_0 = -1$, $\sigma_1 = 1$. Both shear norms are less than two. The sheared map is $(s\mathcal{R}_i, v\mathcal{R}_i/P_i)$. For $g_i = \mathbf{A}_i^{-1}\tilde{F}_i - \text{id}$,

$$J_i g_i = (s\alpha_i, v\omega_i), \quad \alpha_i = e^{\xi_i} - 1, \quad \omega_i = e^{\xi_i} a_i / P_i - 1.$$

It vanishes outside $B(0, 2)$. On its support $|s| \leq 2$, $|v| \leq 2\sqrt{2}$, and

$$\begin{aligned} |\alpha_i| &\leq e^\mu \mu, & \|\nabla \alpha_i\|_2 &\leq e^\mu \sqrt{2} \mu, \\ |\omega_i| &\leq e^\mu \mu(1 + 1/p_-), & \|\nabla \omega_i\|_2 &\leq 2e^\mu \sqrt{2} \mu(1 + 1/p_-). \end{aligned}$$

Differentiating the two components and applying the inverse shear yields

$$\|Dg_i\|_2 \leq 2e^\mu \mu[(1 + 2\sqrt{2}) + (8 + \sqrt{2})(1 + 1/p_-)] < 200\mu/a_* < 1/5.$$

The final coarse estimate follows from $p_- > a_*/2$, $e^\mu < 3$, and $a_* \leq 1/2$. Thus, for every target b , the equation $x = \mathbf{A}_i^{-1}b - g_i(x)$ has a unique solution by contraction on the complete plane. Also $D\tilde{F}_i$ is nonsingular. This proves a smooth global inverse. The zero perturbation values give the required first derivatives.

On Q , $\kappa = 1$ and the angular maps are $Y = \sigma_i + (y - \sigma_i)/P_i(s, sy)$. For $f_s(y) = y - 2\eta(s, sy)$ and $\vartheta = a_0 - a_1$, define $\vartheta_\pm$ by $f_s(\vartheta_\pm) = \vartheta \pm 2us$. The derivative of f_s lies in $[1 - 2\nu s, 1 + 2\nu s]$. The roots are inside $(-1, 1)$ because $0 < P_i < 1$ at the endpoints. The complete feasible domains are $[-1, \vartheta_+]$ and $[\vartheta_-, 1]$, and

$$\frac{4us}{1 + 2\nu s} \leq \vartheta_+ - \vartheta_- \leq \frac{4us}{1 - 2\nu s}.$$

The radial factor is less than one, so there is no cap violation. The domains cover the whole fibre, including the vertex. Theorems 2.3 and 2.5 now apply. $\square$

8.2. Inward boundary images and empty fixed-radius slices. Let $t = \max a_i^{\beta-1} < 1$, $\Lambda = 1 + 4t/(1 - t)$, $v = t(1 + 1/\Lambda) < 1$, and fix

$$0 < c < \min\{1 - a^*, (1 - v)/v, 1\}, \quad g(s) = 1 - \frac{cs}{1 + s^2}.$$

Consider the single explicit family

$$(8.2) \quad \begin{aligned} F_0(s, z) &= (r_0 s, g(s)t_0(z + a_1 s)), \\ F_1(s, z) &= (r_1 s, g(s)t_1(z - a_0 s)). \end{aligned}$$

Both are global smooth diffeomorphisms: $g \in [1 - c/2, 1 + c/2]$ is positive, the first coordinate has the explicit inverse $s = s'/r_i$, and the second is an affine invertible function of z at that radius. Their first jets are (2.3). For the stated V ,

$$(8.3) \quad V(F_i x) \leq \max\{r^*, (1 + c/2)v\} V(x), \quad \max\{r^*, (1 + c/2)v\} < 1.$$

Thus the norm inequality is checked globally, not inferred from feasible trajectories.

For $0 < s \leq 1$, $a^* < g(s) < 1$. The complete one-step angular domains are

$$(8.4) \quad D_0(s) = [-1, a_0/g(s) - a_1], \quad D_1(s) = [a_0 - a_1/g(s), 1].$$

Their endpoints lie strictly inside $(-1, 1)$, and their overlap has angular width $1/g(s) - 1 > 0$, physical width $s(1/g(s) - 1) = O(cs^2)$ near zero. This proves controlled invariance and the actual overlap geometry. Theorem 2.3 applies.

Yet for every fixed $0 < s_0 \leq 1$, a sufficiently long all-zero word is infeasible for every initial angle. To verify this, write $s_k = r_0^k s_0$ and $q_k = y_k + 1$. Under zero input,

$$(8.5) \quad q_{k+1} = \frac{g(s_k)}{a_0} q_k + 1 - g(s_k).$$

The multiplier is uniformly greater than one, since $g(s) > a^*$, and $q_1 \geq 1 - g(s_0) > 0$ for $y_0 \geq -1$. The sequence eventually exceeds two, violating $y \leq 1$. Monotonicity means that this also holds for the smallest initial angle, hence for all of them. Thus a description in which every word has a nonempty full-image slice does not apply to this example.

This statement concerns $I_{0^m}(s_0)$ at a fixed radius, not the whole physical region $\mathcal{I}_{0^m}$. Choosing the radius smaller can make a given finite word feasible. Indeed any finite word admits a bounded-run infinite extension, and Lemma 5.1 realises that extension at a sufficiently small positive radius. Missing fixed-radius slices must not be confused with removing a word everywhere in Q .

There is also a constraint-preserving conjugacy obstruction. For (8.2), F_0 sends the positive lower edge of Q strictly inside Q , and its upper edge outside; F_1 does the symmetric thing. The image of the top edge has constant radius $r_i < 1$ and crosses the opposite angular boundary exactly once. Thus

$$(8.6) \quad F_i(\partial Q) \cap \partial Q \text{ consists of exactly two points.}$$

For every pair in (8.1), control zero preserves the whole lower edge as a boundary continuum and control one preserves the whole upper edge. A homeomorphism H of the plane with $H(Q) = Q$, simultaneously conjugating the two labelled controls (even allowing a label permutation), would map each intersection in (8.6) onto the corresponding target intersection. Finite versus continuum is impossible. Thus this example is not reducible to the class (8.1) by a fixed Q -preserving simultaneous control homeomorphism, and in particular not by such a C^1 or bi-Lipschitz map. No assertion about unconstrained coordinate changes or every member of the new class is made.

For example, take $a_0 = 1/3$, $a_1 = 2/3$, $\beta = 2$ and $c = 1/100$. The choice $\Lambda = 9$ in (8.3) gives a global factor at most $67/90 < 1$. Corollary 2.4 evaluates this same inward family at exponent $\gamma = \log 4$:

$$2 \left(\log 3 - \frac{2}{3} \log 2 \right) < \log 4 < \log(9/2), \quad \lim_n n^{-1} \log r_F(n, \delta_n; Q) = H_b \left(\frac{\log(4/3)}{\log 2} \right).$$

This holds for every precision sequence of that exponent. The first inequality follows from $(4/3)^3 > 2$ and the second from $4 < 9/2$. It is an evaluation of the spectrum, not a claim that missing fixed-radius words require a different count.

8.3. Angular feedback in an admissible mismatched pair. Choose unequal β_0, β_1 in (2.2), and let κ be the fixed cutoff in Section 8.1. Put $\phi(s, z) = z^2 \kappa(s, z)$, $M_\phi = \max\{1, \|x \cdot \nabla \phi\|_\infty\}$, and fix $0 < c < 1/(2M_\phi)$. Define

$$H(x) = e^{-c\phi(x)}x, \quad F_i(x) = A_i H(x).$$

On each ray $x = te$, $t \geq 0$, the image radius has derivative $e^{-c\phi(te)}[1 - cte \cdot \nabla \phi(te)] > 0$ and tends to infinity. Thus H is bijective. Its derivative has determinant $e^{-2c\phi(x)}[1 - cx \cdot \nabla \phi(x)] > 0$; the inverse function theorem gives a smooth global inverse, including at zero. The map is the identity outside a compact set and $DH(0) = I$. Consequently $DF_i(0) = A_i$. Since the scalar factor is at most one, the linear norm inequality checked in Section 2 proves (2.4) globally. The angles are the reference maps (3.1), their domains cover each fibre, and the radius decreases. This proves controlled feasibility with all boundary states retained. On Q , $\phi = z^2$, so the first coordinate $r_i s e^{-cz^2}$ depends on the initial angle at fixed positive radius. Theorem 2.2 applies. This example verifies actual radial feedback; no assertion that it excludes every simultaneous coordinate reduction is made.

9. WHY LOCAL MATCHING ALONE IS INSUFFICIENT

The following example keeps the common attracting origin and even agrees with the linear maps on a neighbourhood of it. It violates global injectivity by collapsing positive-area regions onto an invariant ray.

Proposition 9.1. *For every matched choice $\beta_0 = \beta_1 = \beta$ in (2.2) and every $0 < \sigma < 1/2$, there are C^∞ maps $S_0, S_1 : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ with these properties:*

- (1) *They equal A_0, A_1 on $\{s \leq \sigma\}$, so all derivatives at the origin agree.*
- (2) *All inputs drive bounded sets uniformly exponentially to the origin, and Q is controlled invariant.*
- (3) *The compact set $K_\sigma = Q \cap \{s \geq 2\sigma\}$ has area $1 - 4\sigma^2$ and $r_S(n, \delta; K_\sigma) = 1$ for every n and every $\delta > 0$.*
- (4) *For every finite $\gamma \geq 0$ and every sequence $\delta_n \rightarrow 0$ with $-\log \delta_n/n \rightarrow \gamma$, the full-set cost satisfies*

$$\min\{h, \gamma/\beta\} \leq \liminf \frac{\log r_S(n, \delta_n; Q)}{n} \leq \limsup \frac{\log r_S(n, \delta_n; Q)}{n} \leq \min\{\log 2, \gamma/\beta\}.$$

In particular, for $0 < \gamma \leq \beta h$ the full-set rate is γ/β , but the positive-area set K_σ has rate zero.

Proof. Let $e(t) = 0$ for $t \leq 0$ and $e(t) = \exp(-1/t)$ for $t > 0$, and define

$$\psi(s) = \frac{e(s - \sigma)}{e(s - \sigma) + e(2\sigma - s)}.$$

The denominator is positive everywhere. Thus ψ is smooth, zero for $s \leq \sigma$, and one for $s \geq 2\sigma$. Put $\rho_i = a_i^\beta$ and define

$$(9.1) \quad S_0(s, z) = (\rho_0 s, (1 - \psi(s))a_0^{\beta-1}(z + a_1 s) - \psi(s)\rho_0 s),$$

$$(9.2) \quad S_1(s, z) = (\rho_1 s, (1 - \psi(s))a_1^{\beta-1}(z - a_0 s) + \psi(s)\rho_1 s).$$

They have the claimed local identity. Each $S_i(s, z)$ is a convex combination of $A_i(s, z)$ and $R_i(s, z)$, where $R_0 = (\rho_0 s, -\rho_0 s)$ and $R_1 = (\rho_1 s, \rho_1 s)$. All four linear maps contract toward zero in one adapted norm $\max\{M|s|, |z|\}$ for sufficiently large M : the second-coordinate factor for A_i is at most $a_i^{\beta-1} + a_i^{\beta-1}a_{1-i}/M < 1$, and R_i has factor at most ρ_i . Convexity of the norm gives the same contraction bound for S_i on all of $\mathbb{R}^2$.

For controlled feasibility let $t = \psi(s)$, $y = z/s$, and $\xi = a_0 - a_1$. If $t < 1$ the full feasible intervals are

$$J_0(s) = \left[-1, \min \left\{ 1, \xi + \frac{2a_0 t}{1-t} \right\} \right],$$

$$J_1(s) = \left[\max \left\{ -1, \xi - \frac{2a_1 t}{1-t} \right\}, 1 \right].$$

They follow by imposing $-1 \leq z'/s' \leq 1$ in (9.1)–(9.2), and their union is $[-1, 1]$. For $t = 1$ both branches are feasible on the whole fibre. The radius always decreases, so Q is controlled invariant.

On K_σ , one application of S_0 sends every state to the lower ray. At every radius $S_0(s, -s) = (\rho_0 s, -\rho_0 s)$, so input 0^∞ keeps the whole set exactly in Q . Its area is $\int_{2\sigma}^1 2s ds = 1 - 4\sigma^2$.

To obtain the full-set lower bound, use $Q_c = Q \cap \{s \leq c\}$ with $c = \sigma/2$. For every input, the radius of a state in Q_c stays at most c . Thus $\psi = 0$ forever along that trajectory, even when its angle leaves $[-1, 1]$, and its entire trajectory is exactly that of A_i . Consequently $r_S(n, \delta; Q) \geq r_A(n, \delta; Q_c)$. Corollary 6.2 and (2.10) give the lower bound.

For the upper bound choose an integer t_0 such that $(\max_i \rho_i)^{t_0} \leq \sigma$. Every initial state admits an exact feasible prefix of length t_0 , ending in the region $s \leq \sigma$. All subsequent trajectories there coincide with the linear system, for all inputs, since the radius continues to decrease independently of the angle. Concatenate all 2^{t_0} possible prefixes with a catalogue for the linear full set over the remaining horizon. For $n \geq t_0$ this gives

$$r_S(n, \delta; Q) \leq 2^{t_0} r_A(n - t_0, \delta; Q) \leq 2^{t_0} C_U \delta^{-1/\beta}.$$

The 2^n exact-prefix bound also holds. These yield the upper rates for every stated precision sequence. Finally $\det DS_i = (1 - \psi(s))a_i^{2\beta-1}$ vanishes on the saturated region, identifying the noninjective mechanism. $\square$

For fixed σ this proposition gives one large zero-cost initial set in one fixed system. Making its area arbitrarily close to one changes σ and the system. This differs from Theorem 2.3, where all K_ζ are chosen inside a single fixed system. The collapse example is a counterexample to sufficiency of local matching and uniform convergence; it is not a counterexample to the diffeomorphism theorem.

10. CONSEQUENCES AND SCOPE

Theorem 2.1 characterises the common low-precision cost in the specified first-order family. Unequal radial powers already separate the full triangle and one nearly full-area compact set at arbitrarily small positive exponents. The necessity statement permits angular feedback in the radius and has the same global hypotheses as the matched result.

Theorem 2.5 is the underlying estimate for actual initial states under every approximately feasible input. Its two products record different features: λ_w controls feasible angular size and P_w controls the physical radius at a first true exit. Typicality of all feasible prefixes handles overlap, and the constraint at that first exit handles later reentry. Finite inverse Jacobians control the transient. No positive length or full image is required of each feasible slice.

The full-set lower bounds use rare actual states at a fixed small positive radius, with a margin against the wrong control. Typical positive-area mass instead gives the near-full-set slope. Under matching, the product identity and classical type counting give the complete spectrum and its two distinct thresholds. The auxiliary witness radius depends on the fixed block length; the near-full initial set precedes all stated precision choices.

The results retain binary planar dynamics, a particular triangular constraint, the displayed first-jet family, controlled feasibility, global invertibility, and contraction toward a common origin. They allow truncated feasible images and vanishing overlap, but do

not treat constant overlap in the limiting angular maps. The collapse example explains why the noncollapse hypothesis cannot simply be omitted. The inward example proves a constraint-preserving conjugacy obstruction only for its displayed target class; no claim of nonreducibility is made for every member. The high-precision spectrum under mismatch and the high-side rates of arbitrary compact initial sets are not classified. No fixed-strategy optimality or new pressure principle is asserted.

Generative AI disclosure. Generative AI was used extensively in developing mathematical arguments, checking proof chains and related literature, drafting and restructuring the English text, and producing the LaTeX source. These activities do not constitute verification by a human author. Author identity, affiliations, authorship approval and the scope of human mathematical review remain to be supplied and confirmed before any submission.

REFERENCES

- [1] Z. Chen and X. Zhong, Invariance complexity and equi-invariability for control systems, *J. Math. Anal. Appl.*, **539** (2024), 128533. doi:[10.1016/j.jmaa.2024.128533](https://doi.org/10.1016/j.jmaa.2024.128533).
- [2] H. Chen, Y. Huang and X. Zhong, Variational principles of invariance entropy dimension, *J. Differential Equations*, **453**, Part 1 (2026), 113819. doi:[10.1016/j.jde.2025.113819](https://doi.org/10.1016/j.jde.2025.113819).
- [3] F. Colonius, Minimal bit rates and entropy for exponential stabilization, *SIAM J. Control Optim.*, **50** (2012), 2988–3010. doi:[10.1137/110829271](https://doi.org/10.1137/110829271).
- [4] F. Colonius and B. Hamzi, Entropy for practical stabilization, *SIAM J. Control Optim.*, **59** (2021), 2195–2222. doi:[10.1137/20M1367775](https://doi.org/10.1137/20M1367775).
- [5] I. Csiszár, The method of types, *IEEE Trans. Inform. Theory*, **44** (1998), 2505–2523. doi:[10.1109/18.720546](https://doi.org/10.1109/18.720546).
- [6] J. E. Hutchinson, Fractals and self-similarity, *Indiana Univ. Math. J.*, **30** (1981), 713–747. doi:[10.1512/iumj.1981.30.30055](https://doi.org/10.1512/iumj.1981.30.30055).
- [7] C. Kawan and A. Da Silva, Invariance entropy for a class of partially hyperbolic sets, *Math. Control Signals Syst.*, **30** (2018), Article 18. doi:[10.1007/s00498-018-0224-2](https://doi.org/10.1007/s00498-018-0224-2).
- [8] C. Kawan, *Invariance Entropy for Deterministic Control Systems: An Introduction*, Lecture Notes in Mathematics, 2089, Springer, 2013. doi:[10.1007/978-3-319-01288-9](https://doi.org/10.1007/978-3-319-01288-9).
- [9] X. Nie, T. Wang and Y. Huang, Measure-theoretic invariance entropy and variational principles for control systems, *J. Differential Equations*, **321** (2022), 318–348. doi:[10.1016/j.jde.2022.03.013](https://doi.org/10.1016/j.jde.2022.03.013).
- [10] T. Wang, Y. Huang and H.-W. Sun, Measure-theoretic invariance entropy for control systems, *SIAM J. Control Optim.*, **57** (2019), 310–333. doi:[10.1137/18M1197862](https://doi.org/10.1137/18M1197862).
- [11] R. Yang, E. Chen, J. Yang and X. Zhou, Bowen’s equations for invariance pressure of control systems, *SIAM J. Control Optim.*, **63** (2025), 1104–1128. doi:[10.1137/23M1607684](https://doi.org/10.1137/23M1607684).